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LLM: Google NotebookLM

Original Prompt

Create an infographic on Markov Chains. It must be detailed yet easy to digest. Include visuals to supplement the lessons for a more complete understanding of the topic.

AI Enhancement Prompt to ChatGPT 5.3 Instant:

You are a master prompt engineer. Improve my prompt to create a superior output. This here is my prompt to notebooklm in infographic generation. Create an infographic on Markov Chains. It must be detailed yet easy to digest. Include visuals to supplement the lessons for a more complete understanding of the topic.

AI Enhanced Prompt

Create a **high-quality infographic** explaining **Markov Chains** for students with basic probability knowledge.

Goals:

- Make it **visually intuitive, logically structured, and easy to follow**
- Balance **conceptual clarity** with **technical depth**

Content Structure (strictly follow this order):

1. **Title Section**
 - Clear title: *"Markov Chains Explained"*
 - One-sentence definition (simple and memorable)
2. **Core Idea**
 - Explain the **Markov Property** (memoryless concept)
 - Include a short real-life analogy (e.g., weather, board games)
3. **Visual Diagram**
 - A **state diagram** with:
 - At least 3 states
 - Arrows labeled with transition probabilities
 - Ensure probabilities from each state sum to 1
4. **Transition Matrix**
 - Show the matrix form of the example

- Brief explanation of how to read it
- 5. **Step-by-Step Example**
 - Demonstrate how to move from one state to another
 - Show at least one multi-step transition
- 6. **Key Concepts Box**
 - States
 - Transition probabilities
 - Matrix representation
 - Steady-state (brief intro only)
- 7. **Why It Matters**
 - 3–4 real-world applications (e.g., search engines, text prediction, finance)
- 8. **Design Instructions**
 - Use clean layout with sections
 - Include icons or simple illustrations for each concept
 - Use color coding for states and transitions
 - Avoid dense paragraphs—use bullets and labels

Tone & Style:

- Clear, concise, and student-friendly
- Avoid unnecessary jargon, but keep terms accurate
- Each section should be understandable in under 20 seconds

Output Format:

- Structured infographic content with labeled sections
- Include descriptions of visuals (so they can be rendered)

Reflection:

During the early weeks when this Lab was in Daigler, I planned to set up an MCP server with Claude desktop and connect it to a AI-integrated business intelligence dashboard software, namely, Qlik. This, however, is probably too complicated for the requirements of this assignment, which is why I used it for creating study-guide infographic via NotebookLM instead.

Now, comparing the outputs between the before and after, the one with the AI-enhanced prompt is definitely better.

This activity introduced me to prompt engineering to create ideal instructions that LLMs can use to create outputs that are closer to or beyond the expectations of the user.